

Optimization Workshop: Test functions

1. Paraboloid Function

$$f(\mathbf{x}) = f(x_1, \dots, x_n) = \sum_{i=1}^n x_i^2$$

Search domain: $-5 \leq x_i \leq 5$
Minimum at $\mathbf{x}^* = (0, 0, \dots, 0)$
Min. value $f(\mathbf{x}^*) = 0$

- ⇒ First try with $n = 2$
- ⇒ Then try with $n = 20$

2. Rosenbrock Function

$$f(\mathbf{x}) = f(x_1, \dots, x_n) = \sum_{i=1}^{n-1} [100(x_{i+1} - x_i^2)^2 + (x_i - 1)^2]$$

Search domain: $-2.048 \leq x_i \leq 2.048$
Minimum at $\mathbf{x}^* = (1, 1, \dots, 1)$
Min. value $f(\mathbf{x}^*) = 0$

- ⇒ First try with $n = 2$
- ⇒ Then try with $n = 20$

3. Griewank Function

$$f(\mathbf{x}) = f(x_1, \dots, x_n) = 1 + \frac{1}{4000} \sum_{i=1}^n x_i^2 - \prod_{i=1}^n \cos\left(\frac{x_i}{\sqrt{i}}\right)$$

Search domain: $-600 \leq x_i \leq 600$
Minimum at $\mathbf{x}^* = (0, 0, \dots, 0)$
Min. value $f(\mathbf{x}^*) = 0$

- ⇒ First try with $n = 2$
- ⇒ Then try with $n = 20$

4. Schwefel Function

$$f(\mathbf{x}) = f(x_1, \dots, x_n) = 418.8929n - \sum_{i=1}^n x_i \sin \sqrt{|x_i|}$$

Search domain: $-500 \leq x_i \leq 500$

Minimum at $\mathbf{x}^* = (\alpha, \alpha, \dots, \alpha), \quad \alpha = 420.9687$

Min. value $f(\mathbf{x}^*) = 0$

⇒ First try with $n = 2$

⇒ Then try with $n = 20$

5. Rastrigin Function

$$f(\mathbf{x}) = f(x_1, \dots, x_n) = 10n + \sum_{i=1}^n [x_i^2 - 10 \cos(2\pi x_i)]$$

Search domain: $-5.12 \leq x_i \leq 5.12$

Minimum at $\mathbf{x}^* = (0, 0, \dots, 0)$

Min. value $f(\mathbf{x}^*) = 0$

⇒ First try with $n = 2$

⇒ Then try with $n = 20$

Note: If $n = 20$ does not work, try with smaller n .

Note: In case $n = 2$, you can also do an animation that shows how the search agents approach the point of minimum with each iteration.